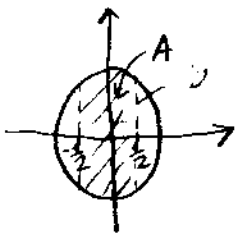
Marmara University
Electrical and Electronics Engineering
Probability and Statistics
Problem Sets With Solutions # 9

[1] For the joint PDF

$$p_{X,Y}(x,y) = \begin{cases} \frac{1}{\pi}, & x^2+y^2 \le 1 \\ 0, & \text{otherwise} \end{cases}$$

find $P[|X| \le \frac{1}{2}]$ { Hint: $\int \sqrt{1-x^2}\,dx = \frac{1}{2}x\sqrt{1-x^2} + \frac{1}{2}\text{Arcsin}(x)$ }

Solution:



$$P[|X| \le \tfrac{1}{2}] = P[X \in A] = \iint_A p_{X,Y}(x,y)\,dy\,dx$$

By symmetry

$$P[|X| \le \tfrac{1}{2}] = 4\,P\left[0 \le x \le \tfrac{1}{2},\ 0 \le y \le \sqrt{1-x^2}\right]$$

$$= 4\int_0^{\frac{1}{2}}\int_0^{\sqrt{1-x^2}} \frac{1}{\pi}\,dy\,dx$$

$$= \frac{4}{\pi}\int_0^{1/2}\sqrt{1-x^2}\,dx = \frac{4}{\pi}\left[\frac{1}{2}x\sqrt{1-x^2} + \frac{1}{2}\text{Arcsin}(x)\Big|_0^{1/2}\right]$$

$$= \frac{4}{\pi}\left[\frac{1}{4}\sqrt{\frac{3}{4}} + \frac{1}{2}\text{Arcsin}\left(\frac{1}{2}\right) - 0 - 0\right] = \frac{\sqrt{3}}{2\pi} + \frac{1}{3}$$

[2] If a joint PDF is given by

$$p_{X,Y}(x,y) = \begin{cases} \frac{c}{\sqrt{xy}}, & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

find c.

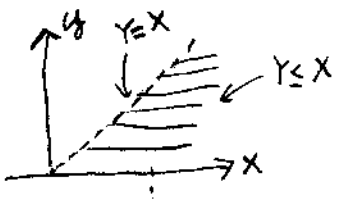
Solution

$$1 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p_{X,Y}(x,y)\,dx\,dy = \int_0^1\int_0^1 \frac{c}{\sqrt{xy}}\,dx\,dy = c\left[\int_0^1 x^{-\frac{1}{2}}dx\right]\left[\int_0^1 y^{-\frac{1}{2}}dy\right]$$

$$= c\left[2\sqrt{x}\Big|_0^1\right]\left[2\sqrt{y}\Big|_0^1\right] = 4c \Rightarrow c = \frac{1}{4}$$

[3] For the joint PDF $p_{X,Y}(x,y) = e^{-(x+y)}u(x)u(y)$, find $P[Y \le X]$

Solution:



$$P[Y \le X] = \int_0^{\infty}\int_0^{x} e^{-(x+y)}\,dy\,dx$$

$$= \int_0^{\infty} e^{-x}\left[\int_0^x e^{-y}dy\right]dx$$

$$= \int_0^{\infty} e^{-x}\left[1 - e^{-x}\right]dx$$

$$= -e^{-x} + \frac{1}{2}e^{-2x}\Big|_0^{\infty} = 1 - \frac{1}{2} = \frac{1}{2}$$

[4] If $P_{X,Y}(x,y) = e^{-(x+y)}\,u(x)u(y)$, determine the marginal PDFS.

Solution: $P_X(x) = \int_{-\infty}^{\infty} P_{X,Y}(x,y)\,dy = \int_{-\infty}^{\infty} e^{-(x+y)}\,u(x)u(y)\,dy = \int_0^{\infty} e^{-(x+y)}\,dy$

$= e^{-x}\int_0^{\infty} e^{-y}\,dy = e^{-x}[-e^{-y}\Big|_0^{\infty}] = e^{-x}$ for $x \geq 0$.

By symmetry, $P_Y(y) = e^{-y}$ for $y \geq 0$.

[5] If

$$P_{X,Y}(x,y) = \begin{cases} 2, & 0<x<1,\ 0<y<x \\ 0, & \text{otherwise} \end{cases}$$

find the marginal PDFS.

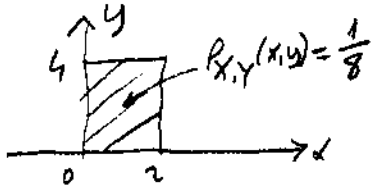
Solution: $P_Y(y) = \int_y^1 P_{X,Y}(x,y)\,dx = \int_y^1 2\,dx = 2(1-y), \quad 0<y<1$

$P_X(x) = \int_0^x P_{X,Y}(x,y)\,dy = \int_0^x 2\,dy = 2x, \quad 0<x<1$

[6] Determine the joint CDF if X and Y are independent with

$$P_X(x) = \begin{cases} \frac{1}{2}, & 0<x<2 \\ 0, & \text{otherwise} \end{cases} \qquad P_Y(y) = \begin{cases} \frac{1}{4}, & 0<y<4 \\ 0, & \text{otherwise} \end{cases}$$

Solution: $P_{X,Y}(x,y) = P_X(x)\,P_Y(y) = \begin{cases} \frac{1}{2}\cdot\frac{1}{4}, & 0<x<2,\ 0<y<4 \\ 0, & \text{otherwise} \end{cases}$

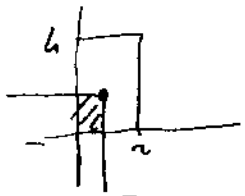


$F_{X,Y}(x,y) = \int_{-\infty}^{y}\int_{-\infty}^{x} P_{X,Y}(t,u)\,dt\,du$

For $x<0$ or $y<0$ $F_{X,Y}(x,y) = 0$

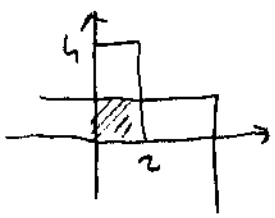
For $x \geq 2$, $y \geq 4$ $F_{X,Y}(x,y) = 1$

otherwise



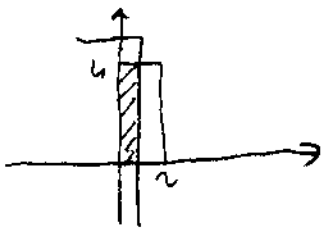
$0 \leq x < 2$
$0 \leq y < 4$

$F_{X,Y}(x,y) = \int_0^y\int_0^x \frac{1}{8}\,dt\,du = \frac{1}{8}xy$



$x \geq 2$
$0 \leq y < 4$

$F_{X,Y}(x,y) = \int_0^y\int_0^2 \frac{1}{8}\,dt\,du = \frac{1}{4}y$



$0 \leq x < 2$
$y \geq 4$

$F_{X,Y}(x,y) = \int_0^4\int_0^x \frac{1}{8}\,dt\,du = \frac{1}{2}x$

In Summary,

$$F_{X,Y}(x,y) = \begin{cases} 0, & x<0 \text{ or } y<0 \\ (1/8)xy, & 0 \leq x < 2,\ 0 \leq y < 4 \\ (1/4)y, & x \geq 2,\ 0 \leq y < 4 \\ (1/2)x, & 0 \leq x < 2,\ y \geq 4 \\ 1, & x \geq 2,\ y \geq 4 \end{cases}$$

[7] Determine the joint CDF corresponding to the joint PDF

$$P_{X,Y}(x,y) = \begin{cases} xy\, e^{-\frac{1}{2}(x^2+y^2)} & , \ x\geq 0,\ y\geq 0 \\ 0 & , \ \text{otherwise} \end{cases}$$

Solution:

$$F_{X,Y}(x,y) = \int_{-\infty}^{y}\int_{-\infty}^{x} P_{X,Y}(t,u)\,dt\,du = \int_{0}^{y}\int_{0}^{x} tu\, e^{-\frac{1}{2}(t^2+u^2)}\,dt\,du$$

$$= \left[\int_0^y u\, e^{-\frac{1}{2}u^2}\,du\right]\left[\int_0^x t\, e^{-\frac{1}{2}t^2}\,dt\right] = \left(-e^{-\frac{1}{2}u^2}\Big|_0^y\right)\left(-e^{-\frac{1}{2}t^2}\Big|_0^x\right)$$

$$= \left(1-e^{-\frac{1}{2}y^2}\right)\left(1-e^{-\frac{1}{2}x^2}\right), \quad x\geq 0,\ y\geq 0$$

[8] If a joint PDF is given as $P_{X,Y}(x,y) = 6\, e^{-(2x+3y)}\, u(x)u(y)$, are the two random variables independent?

Solution:

$$P_{X,Y}(x,y) = 6\, e^{-(2x+3y)}\, u(x)u(y) = \left[2\, e^{-2x}u(x)\right]\left[3\, e^{-3y}u(y)\right] = P_X(x)P_Y(y)$$

Yes, X and Y are independent.

[9] A joint PDF is uniform over the region $\{(x,y): 0\leq y<x,\ 0\leq x<1\}$. Are X and Y independent?

Solution.

$$P_{X,Y}(x,y) = 2\,[u(x)-u(x-1)]\,[u(y)-u(y-x)]$$

We can not factor $P_{X,Y}(x,y)$ as $P_X(x)\cdot P_Y(y)$. So, X and Y are dependent!

[10] If a joint PDF is given as $P_{X,Y}(x,y) = \left(\frac{1}{4}\right)^2 e^{-\frac{1}{2}(|x|+|y|)}$ for $-\infty<x<\infty,\ -\infty<y<\infty$, find the joint PDF of

$$\begin{bmatrix} W \\ Z \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} X \\ Y \end{bmatrix}.$$

Solution:

$$\left.\begin{aligned} W &= 2X+2Y \\ Z &= 2X+Y \end{aligned}\right\} \Rightarrow \left.\begin{aligned} X &= -\tfrac{1}{2}W+Z \\ Y &= W-Z \end{aligned}\right\} \Rightarrow \begin{bmatrix} X \\ Y \end{bmatrix} = \underbrace{\begin{bmatrix} -\frac{1}{2} & 1 \\ 1 & -1 \end{bmatrix}}_{G^{-1}}\begin{bmatrix} W \\ Z \end{bmatrix}$$

$$\boxed{P_{W,Z}(w,z) = P_{X,Y}\left(X(w,z), Y(w,z)\right)\,|\det(G^{-1})|}$$

$$= \left(\frac{1}{4}\right)^2 e^{-\frac{1}{2}\left(\left|-\frac{1}{2}w+z\right|+|w-z|\right)}\cdot\left|-\frac{1}{2}\right|$$

$$= \frac{1}{32}\, e^{-\frac{1}{2}\left(\left|z-\frac{1}{2}w\right|+|w-z|\right)}, \quad -\infty<w<\infty,\ -\infty<z<\infty$$

Note that $G^{-1} = \begin{bmatrix} \frac{\partial x}{\partial w} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial w} & \frac{\partial y}{\partial z} \end{bmatrix}$

[11] If a joint PDF is given as $P_{X,Y}(x,y) = e^{-(x+y)}\, u(x)u(y)$ find the joint PDF of

$$W = XY$$
$$Z = \frac{Y}{X}$$

Solution: $w = xy$, $z = \frac{y}{x}$ $\Rightarrow$ $x = \sqrt{w/z}$, $y = \sqrt{wz}$

$$G^{-1} = \begin{bmatrix} \frac{\partial x}{\partial w} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial w} & \frac{\partial y}{\partial z} \end{bmatrix} = \begin{bmatrix} \frac{1}{2\sqrt{zw}} & -\frac{1}{2}\frac{\sqrt{w}}{z^{3/2}} \\ \frac{1}{2}\sqrt{\frac{z}{w}} & \frac{1}{2}\sqrt{\frac{w}{z}} \end{bmatrix}$$

$$P_{W,Z}(w,z) = P_{X,Y}(x(w,z), y(w,z)) \cdot |\det(G^{-1})|$$

$$= e^{-\left(\sqrt{\frac{w}{z}} + \sqrt{wz}\right)} \cdot \left| \frac{1}{4z} + \frac{1}{4} \cdot \frac{1}{z} \right|$$

$$= e^{-\left(\sqrt{\frac{w}{z}} + \sqrt{wz}\right)} \cdot \frac{1}{2z}, \quad w \geq 0,\ z \geq 0.$$

[12] If

$$P_{X,Y}(x,y) = \begin{cases} e^{-(x+y)}, & 0 \leq y \leq x,\ x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

find $P_{Y|X}(y|x)$

Solution: $P_{Y|X}(y|x) = \frac{P_{X,Y}(x,y)}{P_X(x)}$

$$P_X(x) = \int_0^x P_{X,Y}(x,y)\,dy = \int_0^x e^{-(x+y)}\,dy = e^{-x}\left[-e^{-y}\Big|_0^x\right] = e^{-x}(1 - e^{-x}) = e^{-x} - e^{-2x}$$

$$P_{Y|X}(y|x) = \frac{e^{-(x+y)}}{e^{-x} - e^{-2x}} = \frac{e^{-y}}{1 - e^{-x}}, \quad 0 \leq y \leq x,\ x \geq 0.$$

[13] Determine $P[Y > \frac{1}{2} | X = 0]$ if the joint PDF is given as

$$P_{X,Y}(x,y) = \begin{cases} 2x, & 0 < x < 1,\ 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Solution: $P[Y > \frac{1}{2} | X = 0] = \int_{1/2}^{\infty} P_{Y|X}(y|0)\,dy$

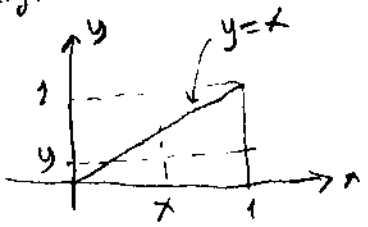
$$P_{Y|X}(y|x) = \frac{P_{X,Y}(x,y)}{P_X(x)}; \quad P_X(x) = \int_0^1 P_{X,Y}(x,y)\,dy = \int_0^1 2x\,dx = 2x$$

$$= \frac{2x}{2x} = 1$$

$$P[Y > \tfrac{1}{2} | X = 0] = \int_{1/2}^{1} 1\,dy = \frac{1}{2}$$

[14] If the joint PDF $P_{X,Y}(x,y)$ is uniform over the region $0 < y < x$ and $0 < x < 1$ and zero otherwise, find the conditional PDFs $P_{Y|X}(y|x)$ and $P_{X|Y}(x|y)$

Solution:



$$P_{X,Y}(x,y) = \begin{cases} 2, & 0 < x < 1 \\ 0 & 0 < y < x \end{cases}$$

$$P_X(x) = \int_0^x P_{X,Y}(x,y)\,dy = \int_0^x 2\,dy = 2y\Big|_0^x = 2x, \qquad 0<x<1$$

$$P_Y(y) = \int_y^1 P_{X,Y}(x,y)\,dx = \int_y^1 2\,dx = 2y\Big|_y^1 = 2(1-y), \qquad 0<y<1$$

$$P_{Y|X}(y|x) = \frac{P_{X,Y}(x,y)}{P_X(x)} = \frac{2}{2x} = \frac{1}{x}, \qquad 0<y<x,\ 0<x<1$$

$$P_{X|Y}(x|y) = \frac{P_{X,Y}(x,y)}{P_Y(y)} = \frac{2}{2(1-y)} = \frac{1}{1-y}, \qquad y<x<1,\ 0<y<1$$

[15] Prove that the minimum mean square error estimate of Y based on $X=x$ is given by $E_{Y|X}(Y|x)$.

Solution: Let the estimate be denoted by $\hat{Y}(x)$ since it will depend on the outcome of X. The mean square error is

$$mse = E_{X,Y}[(Y-\hat{Y}(x))^2] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(y-\hat{y}(x))^2 P_{X,Y}(x,y)\,dx\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(y-\hat{y}(x))^2 P_{Y|X}(y|x)\cdot P_X(x)\,dx\,dy$$

$$= \int_{-\infty}^{\infty}\underbrace{\left[\int_{-\infty}^{\infty}(y-\hat{Y}(x))^2 P_{Y|X}(y|x)\,dy\right]}_{J(\hat{Y}(x))} P_X(x)\,dx$$

Now, we can minimize $J(\hat{Y}(x))$ for each value of x since $P_X(x)\geq 0$. Note that $\hat{Y}(x)$ is a constant since x is assumed fixed.

$$\frac{dJ(\hat{Y}(x))}{d\hat{y}} = \frac{\partial}{\partial\hat{y}}\left[\int_{-\infty}^{\infty}(y-\hat{y}(x))^2 P_{Y|X}(y|x)\,dy\right]$$

$$= \int_{-\infty}^{\infty} -2(y-\hat{y}(x))\,P_{Y|X}(y|x)\,dy = 0$$

$$\Rightarrow \underbrace{\int_{-\infty}^{\infty} y\,P_{Y|X}(y|x)\,dy}_{E_{X|Y}(Y|x)} = \int_{-\infty}^{\infty}\hat{y}(x)\,P_{Y|X}(y|x)\,dy = \hat{y}(x)\underbrace{\int_{-\infty}^{\infty}P_{Y|X}(y|x)\,dy}_{=1}$$

$$\Rightarrow \boxed{\hat{Y}(x) = E_{X|Y}[Y|x]}$$

[16] For the joint PDF given in problem [14], find the minimum mean square error estimate of Y given $X=x$. Plot the region in the x-y plane for which the joint PDF is nonzero and also the estimated value of Y versus x.

Solution: $P_{Y|X}(y|x) = \frac{1}{x} \quad 0 < y < x, \quad 0 < x < 1$

$$\hat{Y} = E_{Y|X}[Y|x] = \int_0^x y\, P_{Y|X}(y|x)\, dy = \int_0^x y \cdot \frac{1}{x}\, dy = \frac{1}{x}\frac{y^2}{2}\Big|_0^x = \frac{x}{2}$$

